

CHESTER JEFFERSON I. ANDAYA

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PROFESSIONAL SUMMARY

Computer Science graduate (Machine Learning), Magna Cum Laude, from National University – Lipa with a 3.67 GPA. Published IEEE conference researcher and builder of end-to-end ML systems using PyTorch, TensorFlow, React, and Django. Founder of an educational community with 50K+ followers across platforms. Seeking AI/ML Engineer, Data Science, or Software Engineering roles.

EDUCATION

B.S. Computer Science — Machine Learning

2022 – 2026

National University – Lipa | Magna Cum Laude | GPA: 3.67 / 4.00 | Dean's Lister

Scholarship: AcadArena Gold Supreme Creator · Achievers Discount Awardee

Relevant Coursework: Machine Learning · Deep Learning · Reinforcement Learning · Data Science · Data Structures & Algorithms · Software Engineering · Operating Systems · Advanced Database Systems · Mathematical Analysis

TECHNICAL SKILLS

ML / AI: PyTorch · TensorFlow · Scikit-learn · CNNs · Reinforcement Learning · Hyperparameter Tuning · Cross-Validation · SHAP · GradCAM

Computer Vision: OpenCV · ResNet50 · MobileNetV2 · Image Augmentation · Multi-class Classification

Data: Pandas · NumPy · Matplotlib · Feature Engineering · Data Cleaning · RapidMiner

Engineering: Python · Java · C++ · REST APIs · Docker · FastAPI · Flask · Django · React · Tailwind CSS

Tools: Git · Postman · VS Code · XAMPP

TECHNICAL PROJECTS

NPK Deficiency Detection in Bitter Gourd Leaves | ResNet50 · PyTorch · OpenCV · React · Django · Docker

Published research · ICMCR 2026, Tokyo, Japan

- ▶ Trained a ResNet50 CNN for 4-class leaf deficiency detection (Healthy / N / P / K) on a 2,500-image dataset validated by the Department of Agriculture Region IV-A; achieved 95% accuracy and 96% F1-score.
- ▶ Applied class-balancing, augmentation, and Bayesian hyperparameter search to close a 7-point accuracy gap versus the baseline.
- ▶ Shipped a full-stack web app (React + Django) supporting real-time image inference and confidence visualisation via GradCAM overlays.
- ▶ Published author and presenter at the International Conference on Machine Learning and Computer Research (ICMCR 2026), Tokyo, Japan.

Enrollment Probability Prediction System | KNN · Random Forest · Scikit-learn · React · Flask

Special Project · NU LIPA 2026

- ▶ Engineered a university admissions analytics platform trained on 20,000+ historical student records; improved predictive accuracy from 83% → 92.67% through feature engineering, stratified cross-validation, and ensemble stacking.
- ▶ Built an interactive analytics dashboard (React + Flask REST API) enabling administrators to drill into cohort-level admission probabilities.

Barako Sense — Liberica Coffee Tree Identification | MobileNetV2 · Multi-Input CNN · React · Django

Cross-institutional collaboration: De La Salle University Manila · UP Los Baños · Batangas State University

- ▶ Designed the first AI-assisted sensory lexicon for Kapeng Barako (Liberica); led model development for a multi-input CNN pipeline achieving 95% accuracy on 4,000+ images from 5 DNA-verified trees.
- ▶ Fused CNN visual features with 100+ expert sensory evaluation records, enabling AI-supported flavour profiling across Liberica cultivars.
- ▶ Delivered a full-stack research platform (React + Django) to help local farmers for tree identification and sensory data visualisation.

- ▶ Extended and maintained CRIS, a live internal platform used by CHEDRO IV-A to manage educational reporting and administrative workflows across higher-education institutions in Region IV-A.
- ▶ Engineered frontend and backend features using React and Laravel, including REST API integrations and database-driven modules that improved data retrieval and operational reporting accuracy.
- ▶ Operated in a production government environment with strict maintainability and deployment standards — applying debugging workflows, code review practices, and usability-focused UI improvements.

WORK & LEADERSHIP

- Full-Stack Developer Intern** · CHED Regional Office IV-A (CHEDRO IV-A) 2026 – July 2026
- ▶ Developed and maintained full-stack features for CRIS, a production information system serving regional higher-education administration across Region IV-A's accredited institutions.
 - ▶ Built REST API endpoints, optimised database queries, and delivered frontend components under real-world production constraints — collaborating directly with senior developers and government supervisors.
 - ▶ Applied professional software engineering practices including version control, code review, systematic debugging, and deployment workflows in a government-regulated environment.
- Founder & Lead Content Creator** · Self-employed (Multi-platform) 2022 – Present
- ▶ Built and grew an educational community to 50,000+ followers across YouTube, TikTok, and Facebook, producing structured ML and programming content targeting Filipino undergraduates.
 - ▶ Designed curriculum-style video series on Python, machine learning fundamentals, and practical deep learning—averaging [X]K views per video.
- Operations Manager & Owner** · Lipa Computer Shop 2022 – 2025
- ▶ Managed day-to-day operations, staff scheduling, inventory, and client relations for a small technology-services business.
- Team Captain & In-Game Leader** · NU–Lipa Esports (Valorant) 2022 – 2025
- ▶ Led a competitive collegiate Valorant roster; responsible for strategy preparation, real-time in-game decision-making, and player development.

PUBLICATIONS & PRESENTATIONS

Andaya, C. J. et al. (2026). *NPK Deficiency Detection in Bitter Gourd Leaves Using ResNet50 CNN* International Conference on Mechatronics, Control and Robotics (ICMCR 2026), Tokyo, Japan.

CERTIFICATIONS

Team Python Specialist
IC3 Digital Literacy